FARIS MUNIR MAHDI

Software Engineer | Backend & Systems Architecture

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PROFESSIONAL SUMMARY

Software Engineer focused on backend and systems engineering, with production experience in meeting-processing and client-delivered IoT applications. Builds Rust and TypeScript services, PostgreSQL data models, and real-time web and mobile integrations. Works across software, infrastructure, security, and UI/UX teams to deliver observable applications while keeping service ownership and authorization boundaries explicit.

SKILLS

Programming	Rust, Go, TypeScript, JavaScript, Python, C++, PHP, Kotlin, SQL, Bash, QML
Web & Mobile	Svelte, Vue, Nuxt, React, HTML, CSS, Android
Backend	Axum, Actix-web, NestJS, Express, FastAPI, Hapi, REST, WebSockets, MQTT, Object-Oriented Design
Data & Storage	PostgreSQL, MySQL, Redis, TimescaleDB, pgvector
Infrastructure	Docker, Linux, GitHub Actions, OpenTelemetry, Prometheus, Grafana
AI Systems	LangGraph, RAG, MCP, Ollama

PROFESSIONAL EXPERIENCE

Backend & Fullstack Developer | PT Perkasa Pilar Utama | July 2025 — Present | Jakarta, Indonesia

Develops backend services, client integration, and production operations for meeting-processing and IoT products, with substantial application ownership inside a cross-functional product team spanning AI services, IoT hardware, infrastructure, security, and UI/UX.

Related projects: Sensio Notes: Meeting Intelligence Platform, Sensio IoT: Smart-Space Platform

- Built the Sensio Notes backend in NestJS with PostgreSQL models, chunked S3 uploads, processing callbacks, and WebSocket updates. Integrated team-owned ingestion and Graph/RAG services into the meeting lifecycle used by web and mobile clients.
- Built major Sensio IoT components in NestJS and Rust/Axum for identity, room state, telemetry, and device integration. Enforced site- and room-scoped authorization before commands and connected MQTT and Zigbee2MQTT to application state.
- Implemented React and Capacitor interfaces from UI/UX designs, handling recording, upload, processing, and recovery states. Collaborated with IoT engineers so software behavior matched device installation and provisioning.
- Packaged services with Docker and diagnosed failures using OpenTelemetry and structured logs. Coordinated with infrastructure and security engineers on on-premises delivery into client-owned residential and office environments.

SEO Specialist | PT Tradeasia International Indonesia | Jan 2024 — Mar 2024

Worked on search visibility through keyword research, on-page SEO, technical checks, and traffic analysis.

- Researched search queries, reviewed page structure and technical SEO issues, and used analytics data to decide which pages and topics needed attention.

SELECTED PROJECTS

Masih Awam AI Code: Agentic Coding Workspace | 2026

Built a coding workspace that connects repository inspection, file changes, command execution, and verification through bounded capabilities. Separated agent reasoning from execution authority with Rust, MCP, and Bubblewrap so repository operations pass through explicit policy boundaries, while the workspace exposes progress and evidence from the actual development environment.

Masih Awam LMS: Game-Based Learning Platform | 2026

Built Rust and Axum services backed by PostgreSQL for courses, enrollment, and lesson completion. Separated auditable progress state from worlds, quests, and story presentation, allowing learner dashboards to show the next step while public course pages and private progress views follow different data ownership and rendering boundaries.

IoTNet: IoT Operations Platform | 2026

Built an IoT management platform using TypeScript, Hapi, PostgreSQL, MQTT, and EMQX to connect user workflows with asynchronous device communication. Separated requested commands from observed device state and checked ownership before control, giving the application a clear model for device operations, automation decisions, and operational visibility through OpenTelemetry.

Multi-Tenant User Management Service | 2026

Built a reusable identity service in Rust and Actix-web with PostgreSQL, JWT sessions, and Argon2 password hashing. Separated global accounts from tenant membership and role assignments so applications can share authentication while evaluating permissions within each tenant, with RocksDB caching and Docker packaging supporting the service runtime.

TECHNICAL PROGRAMS

Machine Learning Track Participant | DBS Foundation | Feb 2025 — July 2025

Related projects: GiziLens: A Team Capstone Around Food Recognition

- Contributed to the application side of the GiziLens team capstone, implementing authentication, sessions, and user APIs around a food-recognition subsystem developed within the team.
- Integrated PostgreSQL and Redis with the Nuxt frontend and containerized services, providing a deployable web application around the team's food-recognition subsystem.

Full Stack Web Development Participant | Codepolitan | Sep 2024 — Dec 2024

- Worked through backend development with Node.js and Express.js, including API design and application data stored in MongoDB.

Backend Engineering Participant | Ruang Guru Academy | Feb 2024 — Jun 2024

Related projects: AskIn: AI Workspace

- Worked through backend-focused learning and implementation around application APIs, service integration, data flow, and deployment.

Cloud Computing Cohort | Bangkit Academy | Aug 2023 — Jan 2024 | Remote · Indonesia

Related projects: Parentify: Parenting Companion

- Completed the Cloud Computing learning path through self-paced coursework, instructor-led sessions, hands-on cloud labs, English practice, and soft-skill training.
- Worked in the Parentify cross-functional capstone team where technical decisions had to be communicated clearly across Cloud Computing, Mobile Development, and Machine Learning roles.

EDUCATION

UPN "Veteran" East Java | Bachelor of Informatics | 2021 — 2025

CERTIFICATIONS

Alibaba Cloud Certified Professional | Alibaba Cloud

BNSP: Junior Web Programmer | LSP UPN Veteran Jawa Timur | Oct 2025

BNSP: IoT Device Engineer | LSP Teknologi Digital | Aug 2026